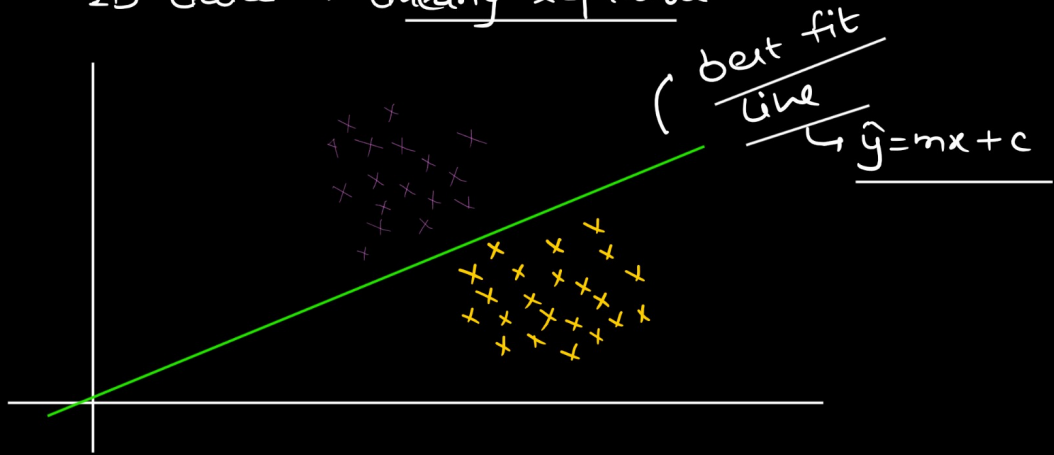
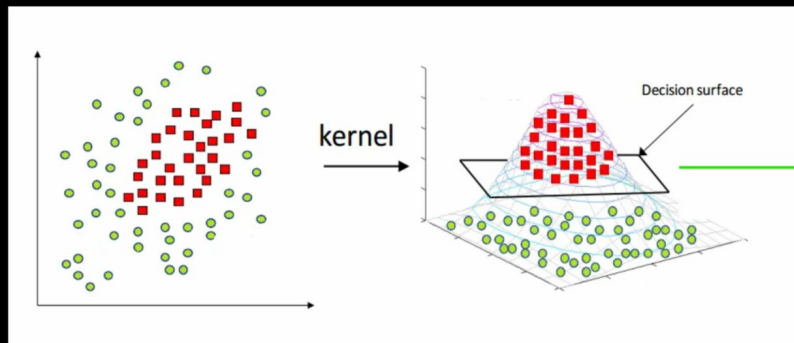


Support Vector Machine

2D data $\rightarrow$ linearly separable



2D $\longrightarrow$ 3D / 4D / 5D (Higher Dimensions)



Hypersurface

$$y = mx + c$$

$$ax + by + c = 0$$

for 2D data,

(line) $w_1x_1 + w_2x_2 + b = 0$

for 3D data,

(plane) $w_1x_1 + w_2x_2 + w_3x_3 + b = 0$

for 4D data,

$$w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + b = 0$$

$$by = -ax - c$$

$$y = \boxed{\frac{-a}{b}}x \quad \boxed{\frac{-c}{b}}$$

slope $m = \frac{-a}{b}$

intercept $c = \frac{-c}{b}$

$$y = mx + c$$

(Hyperplane)

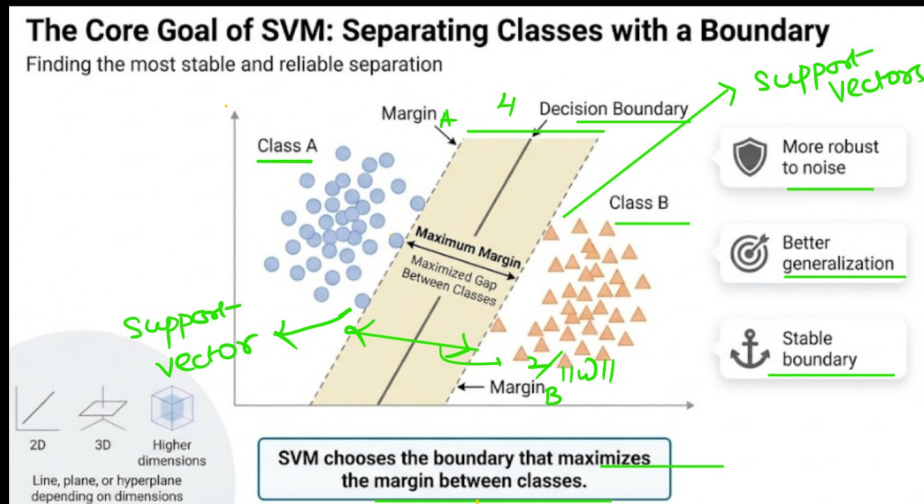
for n D data

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + w_n x_n + b = 0$$

$$\begin{bmatrix} w_1 & w_2 & w_3 & \dots & w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + b = 0$$

$$w^T \cdot x + b = 0$$

class A
test data = 0



Margin line
① $w^T x + b$
② support vectors

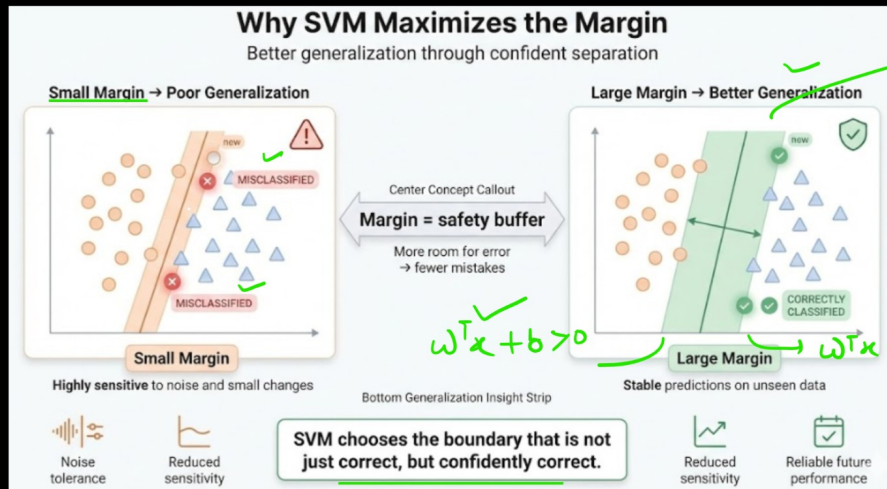
Cost function / $\min$ Error
↓
Optimizer

maximizing the margin b/w classes

③ Decision boundary in SVM

- ① dataset size is small-medium range
- ② num of features is large
- ③ classes are reasonably separable

small margin $\rightarrow$ Poor generalisation



$\checkmark$
 $\omega^T x + b = 0$

$\checkmark$
 $\omega^T x + b > 0$

$\checkmark$
 $\omega^T x + b < 0$

Large margin $\Rightarrow$ Better generalization

optimizers $\rightarrow$ max diff b/w the two
margin

hyperplane $\rightarrow \omega^T x + b = 0$

two regions based on two diff classes

$$\begin{cases} \omega^T x_1 + b > 0 \text{ (one class)} \rightarrow +ve \text{ class} \\ \omega^T x_2 + b < 0 \text{ (another class)} \rightarrow -ve \text{ class} \end{cases}$$

(Two parallel boundaries)

$$\left. \begin{aligned} \omega^T x_1 + b &= +1 \rightarrow +ve \\ \omega^T x_2 + b &= -1 \rightarrow -ve \end{aligned} \right\} \begin{array}{l} \text{Points lying on these} \\ \text{two lines are} \\ \text{called as} \end{array}$$

$\omega^T (x_1 - x_2) = 2$ Projection Support vector

Projected onto unit vector

Margin

$$\hat{\omega} = \frac{\omega}{\|\omega\|} = \frac{\omega^T (x_1 - x_2)}{\|\omega\|} = \frac{2}{\|\omega\|}$$

Width

↳ Maximize the marginal

Width to get stable classifier

Optimizers ← SVM optimization effectively works

Robust,

generalized

